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CASE REPORT

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A giant anterior mediastinal thymolipoma mimicking dextrocardia with cardiomegaly: a case report and review of literature

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Abstract

Background Thymolipoma is a rare benign anterior mediastinal tumour composed of thymic and adipose tissue, accounting for only 2–9% of thymic neoplasms (Mohamud et al., J Surg Case Rep 2020,2020; Shrivastava and Ntiamoah, Radiol Case Rep 15:1538-1540, 2020). Patients are often asymptomatic, and these masses are often detected incidentally. We present a case of a young male with an anterior mediastinal thymolipoma that initially mimicked dextrocardia and cardiomegaly based on findings from clinical examination and chest radiography.

Case presentation A 22-year-old university student was incidentally noted to have abnormal findings on cardiorespiratory examination. A chest X-ray suggested dextrocardia with cardiomegaly, but further imaging revealed a large anterior mediastinal mass. Contrast-enhanced computed tomography (CT) of the chest showed a well-encapsulated, predominantly fat-density mass (approximately $24 \times 15 \times 12$ cm) in the anterior mediastinum extending into the right hemithorax without invasion of adjacent structures. The lesion caused rightward mediastinal widening but no actual cardiac enlargement, dextroposition or lung collapse. Surgical resection via right mini-thoracotomy was performed. The Gross examination revealed a large, lobulated, encapsulated tumour with a yellow, fatty cut surface and interspersed solid areas. Histopathology confirmed the presence of mature adipose tissue with thymic lobules (cortex, medulla, and Hassall's corpuscles), consistent with thymolipoma, showing no evidence of malignancy. The patient's postoperative recovery was uneventful, and the heart resumed its normal anatomical position.

Conclusion This case highlights that an anterior mediastinal mass, such as a thymolipoma, can masquerade as dextrocardia and cardiomegaly by distorting the mediastinal silhouette. It is essential to distinguish a mediastinal tumour from true congenital dextrocardia or pulmonary conditions, as each requires markedly different management approaches. Thorough imaging and histological evaluation ensured the correct diagnosis and curative surgical treatment. This report emphasises the importance of considering mediastinal pathology in patients with apparent dextrocardia, as detected during examination or imaging.

Keywords Thymolipoma, Anterior mediastinal mass, Dextrocardia, Dextroposition, Cardiomegaly, Thymic neoplasm, Case report

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Background

Mediastinal masses are rare but represent clinically significant causes of atypical cardiovascular and pulmonary findings observed during physical examinations and imaging studies. Thymolipoma is a rare anterior mediastinal neoplasm comprising about 2–9% of all thymic tumours [1, 2]. It is a benign tumour that, at times, can grow large and comprises both thymic and fatty tissues. Approximately 200 cases have been documented in the literature [1, 2], and it is most often present in young adults with no apparent gender bias [1]. Nearly all symptomatic cases (30–50%) are discovered through imaging done for unrelated reasons [1, 3]. When present, symptoms typically arise from the mass effect on adjacent structures, leading to nonspecific respiratory symptoms such as cough, chest pain, or dyspnea [4, 5].

An anterior mediastinal mass can occasionally present with unusual clinical signs. In rare cases, a large anterior mediastinal mass may displace mediastinal structures, mimicking dextrocardia or cardiomegaly on imaging. It is important to differentiate true congenital dextrocardia—usually a benign anatomical variant—from dextroposition, where the heart is displaced due to external pathology such as tumours or lung collapse. This distinction has important management implications [3].

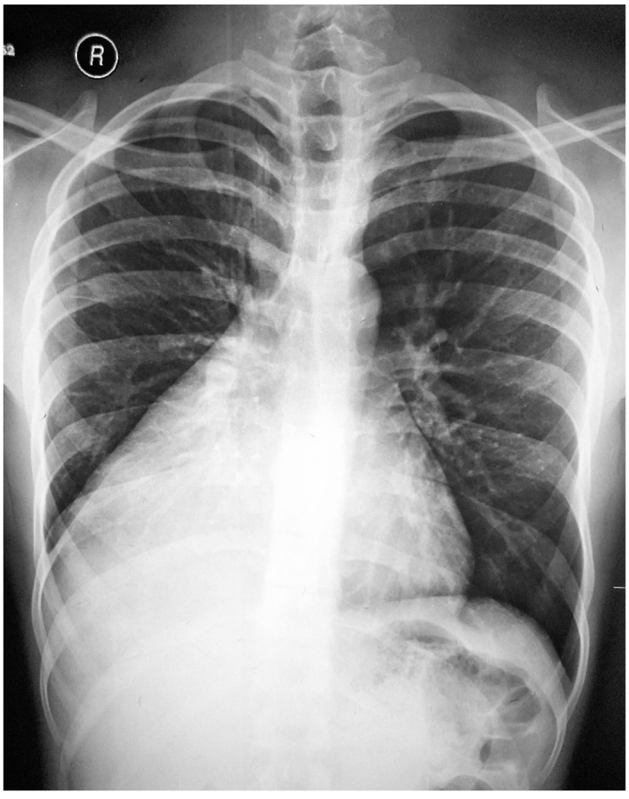


Fig. 1 A postero-anterior chest X-ray mimicking dextrocardia and cardiomegaly

This report presents a case of an anterior mediastinal thymolipoma masquerading as dextrocardia and cardiomegaly in a young adult. The case underlines the importance of a thorough evaluation of apparent dextrocardia, the role of imaging in identifying mediastinal masses, and the need to differentiate such masses from primary cardiac or pulmonary conditions to guide proper management.

Case presentation

Patient information

A 22-year-old university student with no significant past medical history was referred to a tertiary hospital for evaluation of suspected dextrocardia found on a routine examination. There were no significant cardiopulmonary or constitutional symptoms. He did not have a past or family history of cardiac or pulmonary diseases.

Clinical findings

On examination, vital signs were within normal limits. The apex beat could not readily be palpated on the left hemithorax, and heart sounds were feebly audible. Neither the apex beat, nor the heart sounds were clearly identified on the right hemithorax. There were no murmurs. On the respiratory exam, breath sounds were present bilaterally. However, breath sounds were subtly reduced in the right lower anterior lung field. There were no obvious rhonchi or crepitations. The percussion was resonant throughout the lungs except for a slight dullness in the right mid and lower zones. No cervical or supraclavicular lymphadenopathy was appreciated. There were no signs of venous congestion or superior vena cava syndrome.

Diagnostic assessment

An electrocardiogram (ECG) was normal, showing a normal axis and no electrical evidence of dextrocardia (e.g. no global inversion of complexes in lead I and normal R-wave progression). Transthoracic echocardiography revealed normal cardiac anatomy in terms of chambers and orientation, with the cardiac apex pointing leftward (arguing against true situs inversus or dextrocardia) and no evidence of intrinsic cardiac pathology or pericardial effusion. A posterior-anterior chest radiograph, however, demonstrated a rightward shift of the mediastinal shadow, with the cardiac silhouette appearing predominantly in the right hemithorax, raising concern for dextrocardia with cardiomegaly (Fig. 1). Lung fields were clear, with no overt pulmonary lesions, although the mediastinum appeared widened. Further imaging was pursued due to the discrepancy between clinical and ECG findings and the chest X-ray.

The patient was referred to the tertiary care hospital for further evaluation. However, a careful review of the chest

X-ray revealed an abnormal contour on the right side of the mediastinum, suggesting the possibility of a mediastinal mass mimicking cardiomegaly and dextrocardia, or a mass causing dextroposition of the enlarged heart. In true dextrocardia (particularly with situs inversus), one would expect associated findings such as a right-sided gastric air bubble or liver on the left, which were not observed.

A contrast-enhanced CT of the chest was performed for further mediastinal evaluation. Imaging (Fig. 2A, B, C) demonstrated a large, lobulated anterior mediastinal mass extending into the right pleural space and measuring approximately $11.5 \times 9.3 \times 22.7$ cm in anteroposterior, transverse, and craniocaudal dimensions, respectively. The lesion was predominantly of fat density, interspersed with internal soft tissue septations and nodular components, but without calcifications. The mass was well-encapsulated and in close contact with the pericardium along its anterior and right lateral margins, extending inferiorly to the right cardiophrenic angle — a recognised subregion of the anterior mediastinum and abutting the right hemidiaphragm. Contrast-enhanced images (Fig. 2B) revealed minimal enhancement of septa and soft-tissue elements, with no solid-enhancing areas. It exerted mild compressive effects on adjacent structures, particularly the right middle lobe, without evidence of overt atelectasis, collapse, or invasion into the pericardium or great vessels. Surrounding fat planes remained intact, and no mediastinal lymphadenopathy was noted. Both lungs were free of nodules or masses, and no pleural effusion was observed. The heart remained in its normal position despite the presence of the mass.

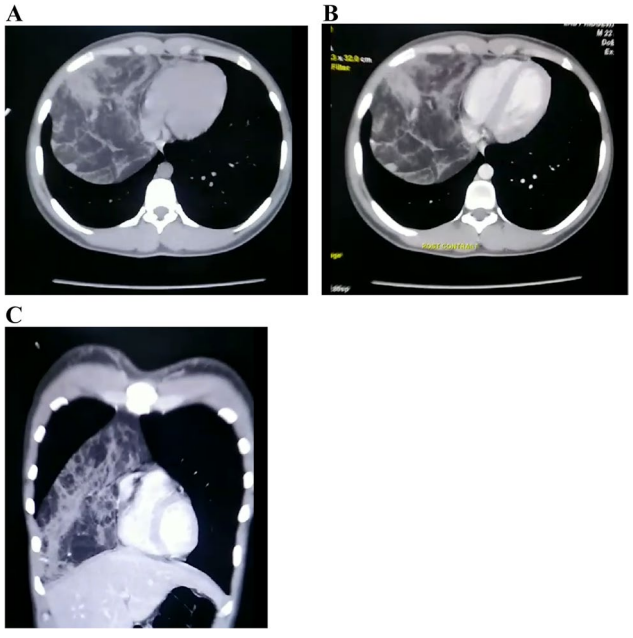


Fig. 2 **A**, **B**: CT chest axial view (**A**: pre-contrast, **B**: post-contrast) showing a large fat-containing structure with internal stranding and nodularity. **C**: Contrast-enhanced CT chest coronal views showing a large mass of fatty tissue with soft tissue density from the anterior mediastinum extending to the right hemidiaphragm

The visualised upper abdominal organs, including the liver, spleen, and adrenal glands, appeared normal in position and morphology. These imaging characteristics—particularly the large, fat-predominant anterior mediastinal mass without invasive features—highly suggest a thymolipoma, a rare benign tumour of thymic origin.

While CT demonstrated a well-encapsulated, fat-dense anterior mediastinal mass with soft tissue strands, definitive attribution to the thymus required histopathological confirmation [3]. The tumour's location in the thymic bed, lack of invasive characteristics, and imaging pattern raised the suspicion of thymolipoma, though differential considerations such as liposarcoma—particularly dedifferentiated or myxoid subtypes—were acknowledged [3]. Liposarcomas typically exhibit irregular margins, heterogeneous density, and enhancement, which were absent in our case. Nevertheless, imaging alone could not confirm the thymic origin; diagnosis was confirmed by histopathology, which demonstrated characteristic thymic elements amidst mature adipose tissue [5].

Therapeutic intervention

Given its size and potential for compression, the patient was subsequently referred for surgical intervention. Through a right mini-thoracotomy, a large encapsulated anterior mediastinal tumour was resected completely. Intraoperatively, the mass was found to be lobulated and loosely adherent to the pericardium; it was successfully dissected free without injury to adjacent structures. The tumour weighed 2.9 kg.

Pathological findings

A gross examination of the excised tumour revealed a lobulated, yellow, fatty, encapsulated mass with fibrous septa. (Fig. 3) The tissue was predominantly adipose on the cut section with interspersed tan-pink solid areas. Histopathological analysis reinforced the diagnosis. Microscopic examination revealed lobules of mature adipose tissue separated by thin fibrous septae. Within and adjacent to these adipose lobules were islands of thymic tissue showing normal architecture, including cortical and medullary differentiation and the presence of Hassall's corpuscles. Occasional foci of hyalinization were present, and some Hassall's corpuscles were calcified. There was no evidence of atypia or malignancy. A rim of residual normal thymus was seen at the periphery of the lesion, outside the capsule, with mild lymphoid hyperplasia. These findings were diagnostic of a thymolipoma.



Fig. 3 Large encapsulated thymolipoma completely resected during surgery

The tumour was confirmed to be completely excised, with clear margins and an intact capsule.

Follow-up and outcomes

The patient had an uneventful postoperative recovery. By the first postoperative week, he remained asymptomatic, and on examination, the apex beat was felt well at the left fifth intercostal space at the midclavicular line. A follow-up chest X-ray one month after surgery showed normalisation of the cardiac silhouette and mediastinal position. At 6-month follow-up, the patient remained symptom-free. He was counselled on the benign nature of his condition and the excellent prognosis after complete resection.

Discussion

In this case, initial chest radiography suggested an enlarged right-sided cardiac silhouette, raising concern for dextrocardia with cardiomegaly. In true dextrocardia, the ECG characteristically shows features such as right-axis deviation, positive QRS complexes in lead aVR with inverted complexes in lead I (a global reversal of the usual pattern), and absent R-wave progression in the precordial leads. Our patient's ECG was normal and did not

support dextrocardia. Transthoracic echocardiography was a crucial bedside tool to evaluate for intrinsic cardiac anomalies and confirmed no cardiomegaly or structural dextrocardia. The contrast-enhanced CT clarified that the apparent dextrocardia and cardiomegaly on the chest X-ray were due to a large anterior mediastinal mass (thymolipoma), causing silhouette overlap. Notably, the CT scan showed that the heart was anatomically located in the left hemithorax, merely displaced and silhouetted by the mass, with no evidence of true cardiac malposition or significant mediastinal shift. This case highlights the importance of cross-sectional imaging in distinguishing true cardiac anomalies from those mimicked by mediastinal masses. Such lesions can obscure the normal cardiac borders on radiographs and simulate cardiomegaly or dextrocardia, emphasising the need for careful correlation with advanced imaging to avoid misdiagnosis [6].

Thymolipomas, while benign, can grow to enormous sizes before detection [5]. In this case, the tumour reached over 20 cm in length, occupying a substantial portion of the right hemithorax and producing a false appearance of dextrocardia and cardiomegaly on initial imaging. True congenital dextrocardia is a rare condition (estimated ~ 1 in 10,000 births) in which the heart is situated on the right side of the chest, usually as part of situs inversus or other congenital anomalies [7]. By contrast, acquired rightward mediastinal shift (sometimes termed dextroposition of the heart) occurs when an external force or pathology displaces an otherwise normal heart to the opposite side [7]. Known causes of such displacement include volume loss of one lung (e.g., post-pneumonectomy or significant lung collapse), large pleural effusions, tension pneumothorax, diaphragmatic hernia, or massive intrathoracic tumours [7]. In our patient, the absence of the classic ECG features of true dextrocardia and the presence of an abnormal mediastinal shadow on imaging prompted further investigation, leading to the diagnosis of an anterior mediastinal mass rather than a primary cardiac malposition.

From a pulmonologist's perspective, differentiating a mediastinal mass from intrinsic lung pathology is critical. Large anterior mediastinal masses may compress airways or lung parenchyma and can be misinterpreted as primary pulmonary pathology on initial imaging. For example, on a chest radiograph, a large right-sided mediastinal mass could obscure the right lung field, potentially mimicking a right pleural effusion or collapse of the left lung with the mediastinal shift. In our case, the chest X-ray showed an unusual right-sided opacity and mediastinal enlargement. One might have considered a large right pleural effusion or left lung collapse causing a shift; however, the clarity of the lung fields and the delineation of the mass on subsequent imaging argued against those diagnoses. Indeed, there are reports of thymolipomas initially being misinterpreted as other conditions on chest X-ray – one case report described a huge thymolipoma simulating bilateral pulmonary oedema on radiography [1]. Clinicians should, therefore, consider a broad differential diagnosis, including mediastinal tumours, when evaluating atypical chest radiographs, particularly when clinical findings do not align with common cardiac or pulmonary conditions. The differential diagnosis of anterior mediastinal masses in young adults is commonly recalled using the classic '5 T's' mnemonic—Thymoma, Thyroid (ectopic or substernal goitre), Terrible lymphoma, Teratoma (and other germ cell tumours), and Thymolipoma—with the addition of liposarcoma as an important malignant consideration. Liposarcoma is a rare malignant tumour of soft tissue origin. Liposarcomas often demonstrate more aggressive features on imaging, such as poorly defined borders, heterogeneity, internal necrosis, and solid-enhancing components [8]. Thymolipoma is one of the least common of these entities. The clinical context, imaging characteristics, and relevant laboratory findings are important considerations in differentiating these diagnoses. Our patient's imaging features – a large, encapsulated fat-density mass with soft-tissue components – highly suggest a thymolipoma (thymic and adipose elements) rather than other anterior mediastinal tumours.

The exact aetiology of thymolipoma remains unclear. It is hypothesised to arise from normal thymic tissue undergoing fatty infiltration or metaplastic transformation, though no definitive molecular or genetic driver has been identified to date. Some studies suggest it may represent a hamartomatous or lipomatous variant of thymic hyperplasia [4, 5].

The prognosis of thymolipoma is excellent. It is a benign lesion with no malignant potential, and complete surgical excision is curative [5]. Recurrence is exceedingly rare, especially when the tumour is resected with intact margins. Unlike other thymic tumours, thymolipomas are not associated with local invasion, metastasis, or aggressive behaviour, making long-term outcomes highly favourable [3].

Thymomas typically present in slightly older patients and, on CT, appear as solid soft-tissue masses (occasionally with calcifications or cystic changes) and can be invasive. Lymphomas of the mediastinum often cause more acute symptoms, may present with lymphadenopathy elsewhere, and appear as homogeneous soft-tissue masses that can encase mediastinal vessels. Germ cell tumours (e.g., mature teratomas) can contain fat and calcification but usually also have cystic areas or even teeth and bone density on imaging; none of these features were seen in this case, making a teratoma less likely. Ectopic thyroid goitres may show iodine uptake on thyroid scans and often demonstrate continuity with the cervical

thyroid gland. A pure mediastinal lipoma (composed only of fat) could resemble a thymolipoma on imaging, but thymolipomas characteristically contain soft-tissue thymic elements [3]. In fact, it can be challenging to distinguish a large thymolipoma from a mediastinal lipoma radiologically. Clues favouring a thymolipoma include its origin in the thymic bed and the presence of internal septations or soft-tissue strands representing thymic tissue [3]. In our case, histopathological examination definitively excluded malignancy, revealing no cellular atypia, mitotic figures, or necrosis typical of liposarcoma. Moreover, the presence of structured thymic lobules with Hassall's corpuscles confirmed thymic origin and benign nature. In our patient, histopathological analysis confirmed the diagnosis, which remains the definitive standard.

It is worth noting the association of thymolipoma with certain systemic conditions. Approximately 50% of thymomas are associated with myasthenia gravis, but thymolipomas have a much weaker association with paraneoplastic syndromes [3]. Occasionally, thymolipomas have been reported in patients with autoimmune or haematologic abnormalities such as hyperthyroidism, aplastic anaemia, or red cell aplasia [1, 3]. Our patient did not have any such associated disorders. The occurrence of thyrotoxicosis in some reported thymolipoma cases is likely incidental [4]. There have also been rare reports of thymolipoma causing immunological abnormalities like hypogammaglobulinemia or, as in one pediatric case, extreme T-cell lymphocytosis [4]. None of these abnormalities were identified in our patient. The management of thymolipoma is straightforward once the diagnosis is made. Complete surgical excision serves both diagnostic and therapeutic purposes. Given the benign nature of the tumour, surgery results in a cure, and recurrence is exceptionally rare when resection is complete [9]. In our patient, a full resection was achieved, and the encapsulated nature of the tumour made removal relatively uncomplicated despite its size. Careful surgical planning is required for large mediastinal tumours to ensure minimal complications; in this case, a right mini-thoracotomy provided adequate exposure. However, an open approach with midline sternotomy may be necessary for a mass of this size. Postoperatively, patients should be followed up with periodic clinical exams and imaging. The long-term prognosis is excellent because thymolipomas do not recur when fully excised [1]. In contrast, had this mass been a malignant tumour (like lymphoma or thymic carcinoma), treatment would have included chemotherapy and/or radiotherapy, and the prognosis and follow-up strategy would have differed greatly. This case underscores the importance of accurate diagnosis and appropriate management.

Intraoperatively, although the mass appeared well-encapsulated and benign, it is important to note that distinguishing thymolipoma from liposarcoma based solely on gross morphology or frozen-section histology can be unreliable [4]. Liposarcomas may require identification of lipoblasts or cytological atypia, not always evident on frozen sections [9]. Therefore, complete surgical resection with clear margins remains the standard approach in managing anterior mediastinal tumours when the exact pathology is not definitively known intraoperatively [10]. In our case, the tumour was excised entirely with intact capsule, and histology later confirmed the diagnosis of benign thymolipoma.

Our case also highlights the importance of interdisciplinary collaboration. The initial clinical assessment by the pulmonology and cardiology teams raised suspicion of an abnormality beyond a congenital heart condition. Radiologists were instrumental in accurately interpreting the chest radiograph and characterising the lesion on CT, effectively distinguishing it from alternative diagnoses, such as a large pericardial cyst or diaphragmatic hernia. Ultimately, thoracic surgeons played a crucial role in providing curative treatment. The favourable patient outcome highlights how recognising an unusual presentation—apparent dextrocardia—can facilitate the diagnosis of a significant but treatable condition.

Conclusion

Patients with apparent dextrocardia should be carefully evaluated to rule out extrinsic causes of cardiac displacement. This case demonstrates that an anterior mediastinal thymolipoma, although rare, can present with clinical and radiographic features that mimic dextrocardia and cardiomegaly. We emphasise that conclusions from a single case have inherent limitations; however, our findings add to the body of evidence on the importance of careful evaluation in apparent dextrocardia. A high index of suspicion and appropriate imaging (such as CT or Magnetic Resonance Imaging) is crucial to differentiating a mediastinal mass from true congenital dextrocardia or other thoracic conditions. Surgical resection of thymolipoma is curative and prevents potential cardiopulmonary complications from mass effect. Early identification and management of such cases result in excellent outcomes. In summary, not all right-sided "hearts" on examination or X-rays represent true dextrocardia – some, as in our patient, are mediastinal masses masquerading as right-sided hearts.

Abbreviations

ECG Electrocardiogram
CT Computed Tomography

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Authors' contributions
NL and UW conceptualized and designed the study, acquired data, analyzed and interpreted data, and drafted the article. MD contributed to designing the study, acquiring data, and analyzing and interpreting data. DR and DS contributed to clinical care, data acquisition, and critical revision of the article for important intellectual content. All authors read and approved the final manuscript.Niroshan Lokunarangoda (NL), Dhammike Rasnayake (DR), Miyuru Dulanjana (MD), Duminda Samarasinghe (DS), Ushani Wariyapperuma (UW).

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Data availability
The patient data supporting the findings of this study are available from the authors. Data are, however, available from the authors upon reasonable request. Contact the corresponding author, Niroshan Lokunarangoda (email: niroshanl@uom.lk).

Declarations

Ethics approval and consent to participate
The treatment was conducted ethically in accordance with the World Medical Association Declaration of Helsinki and the standard of practice in Sri Lanka. The patient provided written informed consent to publish this case report and all the accompanying images.
(This is a case report on patient management. It is not a clinical trial or part of another clinical study. Therefore, registration or formal ethical approval is not needed).

Consent for publication
Written informed consent was obtained from the patient to publish this case report and any accompanying images.

Competing interests
The authors declare no competing interests.

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